Document 10: Data-Driven Expansion Asset & Silent Sync Specification

10.1 The Extensible Pedal Asset Schema (`.pedal`)

To eliminate native code recompilation, expansion pedals are defined strictly as data configurations serialized into a packed binary or JSON format. A `.pedal` bundle contains three essential structural maps:

- **Metadata Header:** Unique 16-bit Asset Identification Token, display name string, and versioning integer.
- **UI Asset Payload:** An array of raw bitmap color matrices mapping directly to the custom pixel-art engine configurations.
- **Parameter Map Descriptor:** An array of structures matching Document 3. Each entry declares a target parameter's token, absolute DSP scale boundaries, and a polymorphic routing tag defining how pixel chunk accumulation translates to processing logic.

10.2 Core Polymorphic DSP Engine

The core VST binary implements a unified, multi-mode DSP block matrix containing universal digital signal building blocks. The processing loop does not handle unique third-party code; instead, it reconfigures its internal DSP routing nodes dynamically based on the parsed `.pedal` schema definitions:

- **Gain & Distortion:** A generic waveshaping array whose saturation curves and clipping thresholds are initialized by coefficients inside the parameter map.
- **Filtering:** A multi-stage cascading biquad filter architecture whose operational mode (Low-pass, High-pass, Band-pass, Notch) is toggled directly by the asset token configuration.
- **Time & Modulation:** A universal delay line and Low-Frequency Oscillator (LFO) matrix with variable-length buffers and flexible interpolation paths.

10.3 Automated Background Sync Pipeline

The synchronization service updates the local asset pool silently within the host DAW without user intervention, manual directory nesting, or global system updates.

1. **The Authorization Handshake:** Upon successful initialization via the serverless routine outlined in Document 8, the cloud endpoint responds with a payload containing an array of authorized asset IDs linked to that specific user account profile (`Entitled_Asset_IDs`).
2. **Local Delta Evaluation:** The plugin executes a local file system scan of its designated local directory (`%APPDATA%/Vendor/Plugin/Assets` on Windows, `~/Library/Application Support/Vendor/Plugin/Assets` on macOS) to generate an inventory array of assets currently residing on disk (`Local_Asset_IDs`). It computes the missing delta matrix via:

$$Delta = Entitled\_Asset\_IDs \setminus Local\_Asset\_IDs$$

3. **Asynchronous Non-Blocking Retrieval:** If the computed delta contains missing items, the Host Compiler Thread initiates sequential, asynchronous secure HTTP GET operations directed at the public Cloudflare KV/D1 file-hosting endpoint.

4. **Thread-Safe Cache Injection:** Downloaded `.pedal` files are saved directly to the local application directory. Once writing completes, the background thread updates a lock-free global registry pointer. The GUI Viewport thread immediately detects this alteration and populates the pedal choice selection list smoothly, making the asset ready for instanced selection without dropping audio blocks or blocking UI responsiveness.